A Textbook-Grounded AI Learning Companion

A concept for curriculum-aligned, source-grounded student support

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1 The idea

Convert a student's semester textbooks and curriculum-aligned supplementary books into a searchable, carefully structured knowledge collection. A learning assistant uses that collection as its first reference when explaining course material, while general knowledge is used selectively to clarify, connect, or extend the curriculum. Answers should make their evidence visible: students can see the source title, chapter, and page or section wherever available.

The system is not simply a chatbot trained on book text. It is a study companion built around the student's actual courses: it helps explain concepts, locate evidence, practise recall, identify gaps, and connect related ideas without replacing teachers, assigned reading, or the student's own reasoning.

2 The student experience

A student selects a course, topic, or uploaded assignment and asks a question in natural language. The assistant searches the course collection, retrieves the most relevant passages, and responds at the requested level of detail. It can offer a plain-language explanation, a worked example, a comparison, a short quiz, or a step-by-step hint. The student can ask follow-up questions, request a simpler explanation, or open the cited passage to verify the answer.

A useful default is "teach, then check": explain the idea briefly, show the textbook evidence, and offer one low-stakes question to test understanding. When the student is doing assessed work, the assistant can guide with hints and reasoning prompts rather than provide a submission-ready answer. The student remains in control of whether supplementary context is included.

3 How the knowledge system works

1. **Collect and organize.** Import authorized textbooks, teacher-provided materials, and approved supplementary works. Keep each item attached to course, semester, edition, language, and rights information. **Extract and preserve structure.** Convert text into machine-readable form while retaining book, chapter, heading, page, table, figure, and equation references. Apply OCR to scanned pages and flag low-confidence regions for review. **Prepare reliable passages.** Clean layout noise without discarding meaning. Break content into coherent, overlapping passages aligned to sections; store provenance and extraction quality with every passage. **Retrieve before responding.** For each question, search within the selected course first. Rank passages for relevance, retrieve a small evidence set, and provide it to the response model. Use supplementary sources only when needed or requested, and clearly label them. **Answer with evidence and limits.** Generate an explanation from the retrieved material, attach citations, and state when the collection does not support a confident answer. Do not invent page references or claim that an uncited statement came from a textbook. **Improve safely.** Use student feedback and evaluation results to improve search and explanation quality. Do not silently turn personal questions, grades, or private conversations into training data.

4 Why retrieval is a better starting point than textbook memorization

The initial idea describes the books as data used to train the AI. For a first product, it is usually more practical to index the books for retrieval and provide relevant passages to the model at answer time. This keeps answers tied to the current edition, makes citations possible, and allows a school to correct, replace, or remove a source without retraining the model. Fine-tuning may later help with a specific response style or teaching routine, but it should not be treated as a substitute for a searchable, auditable source library.

This distinction also supports honest attribution. The model supplies language and reasoning; the retrieved book passages supply course-specific evidence. The assistant should distinguish direct textbook coverage from its own explanatory bridge or from supplementary material.

5 Answer standards

1. **Course-first.** Prefer the selected course's assigned textbook and teacher-approved materials over unrelated web or general model knowledge. **Cited.** Attach citations to the passages that support key claims. Display concise source details and make the cited section easy to open. **Faithful.** Preserve definitions, assumptions, notation, and scope. If sources disagree, show the difference rather than silently blending them. **Calibrated.** Say when evidence is missing, conflicting, poorly scanned, or outside the uploaded collection. Ask a clarifying question when course or topic is unclear. **Educational.** Explain at an appropriate level, encourage active recall, and offer hints before complete solutions when the context suggests practice or assessment. **Transparent.** Label supplementary context and any general explanation that goes beyond the cited course material.

6 Trust, privacy, and responsible use

Textbooks are often copyrighted. Before processing or sharing any book, confirm that the student or institution has the necessary rights and permissions. A student's personal access to a book does not automatically authorize redistribution, public indexing, or use in model training. Keep source files and extracted text private to the authorized learning environment, respect license restrictions, and provide deletion and retention controls.

Collect only the student information needed to deliver the service. Protect course materials and conversation data with appropriate access controls, encryption, retention limits, and clear account boundaries. For younger learners, follow applicable school, guardian, and jurisdictional requirements. Do not use the assistant as the sole authority for grades, high-stakes decisions, or wellbeing advice. Give educators and administrators ways to review source coverage, report errors, and disable unsuitable features.

The assistant should support academic integrity rather than undermine it. Offer learning-oriented modes, disclose AI assistance where required by school policy, and avoid presenting generated work as the student's unaided effort. Build accessibility into the experience: readable language, screen-reader-compatible citations, keyboard navigation, and support for the languages used in the course.

7 Example interaction

Student: "Why does increasing temperature speed up this reaction?"
Assistant: "Your chemistry text explains that particles at a higher temperature have greater average kinetic energy. As a result, a larger fraction of collisions can overcome the activation-energy barrier, so more collisions lead to reaction. The relevant explanation is in *[Course textbook]*, Chapter 6, 'Reaction Rates,' p. 142. Would you like a simple energy-diagram explanation or a quick practice question?"

Illustrative example only. In a real system, the title, chapter, and page must come from retrieved source metadata.

8 A practical rollout

1. **Pilot one course.** Select a single subject and semester, confirm source permissions, and prepare a small set of approved books and course materials. **Build the source library.** Extract and index the content with stable chapter and page metadata. Review OCR quality on scanned pages and test retrieval against questions written by teachers and students. **Launch a limited study assistant.** Provide course-scoped Q&A, citations, explanations, and practice prompts. Make source coverage and limitations visible to users. **Evaluate before expanding.** Measure citation correctness, evidence relevance, factual accuracy, abstention when evidence is missing, student comprehension, accessibility, and teacher confidence. Review errors by topic and source type. **Expand deliberately.** Add courses, languages, and features only when rights, quality, privacy, and support processes are in place. Recheck every new edition and curriculum change.

9 How success can be measured

The pilot should be judged on learning usefulness and trust, not answer fluency alone. Track whether cited passages actually support answers; whether students can find and understand those passages; whether explanations match course level; whether the assistant admits uncertainty appropriately; and whether short quizzes or teacher-reviewed activities show improved understanding. Also monitor time to answer, failed searches, accessibility issues, source-processing errors, and reports of academic-integrity concerns. Establish baseline measures and review them with educators before making claims about learning impact.

10 Closing vision

A well-designed textbook-grounded assistant can make a student's own curriculum easier to navigate, revisit, and understand. Its defining promise is not that it knows everything, but that it can explain course ideas clearly, point back to trustworthy learning materials, and help each student take the next step in their own learning.